CSE330: NUMERICAL METHODS

ASSIGNMENT 01

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Section: 02

Answer to the Question No - 1

(a) Given, $\beta = 2$, $m = 5$, $-2 \leq e \leq 5$

Standard Form

$$\text{Max} = (0.11111)_2 \times 2^5$$

$$= (1\times 2^{-1} + 1\times 2^{-2} + 1\times 2^{-3} + 1\times 2^{-4} + 1\times 2^{-5})_{10} \times 2^5$$

$$= (31)_{10}$$

IEEE Normalized

$$\text{Max} = (0.111111)_2 \times 2^5$$

$$= (31.5)_{10}$$

IEEE Denormalized

$$\text{Max} = (1.11111)_2 \times 2^5$$

$$= (63)_{10}$$

(b) Minimum (Non-Negative),

Standard Form

$$\text{Min} = (0.10000)_2 \times 2^{-2}$$

$$= (0.125)_{10}$$

IEEE Normalized

$$\text{Min} = (0.100000)_2 \times 2^{-2}$$

$$= (0.125)_{10}$$

IEEE Denormalized

$$\text{Min} = (1.00000)_2 \times 2^{-2}$$

$$= (0.25)_{10}$$

(c) Including Negatives numbers,

Standard Form:

Underflow to 0: $\cancel{(0.125)_{10} \leq x \leq (0.125)_{10}} \rightarrow \cancel{|x| \leq |\pm(0.125)_{10}|}$; $\cancel{x \leq |\pm(0.125)_{10}|}$; $\cancel{(-1.25)}$ $(-0.125, 0.125)$

Overflow to $\pm\infty$: $\cancel{-\infty \leq -(31)_{10} \geq x \geq (31)_{10}} \rightarrow \cancel{|x| \geq (31)_{10}}$; $\cancel{x \geq |\pm(31)_{10}|}$

$+\infty \rightarrow (31, \infty)$

$-\infty \rightarrow (-\infty, -31)$

IEEE Normalized Form:

Underflow to 0: $\cancel{x \leq |\pm(0.125)_{10}|}$; $(-0.125, 0.125)$

Overflow to $\pm\infty$: $\cancel{x \geq |\pm(31.5)_{10}|}$; $+\infty \rightarrow (31.5, \infty)$

$-\infty \rightarrow (-\infty, -31.5)$

IEEE Denormalized Form:

Underflow to 0: $\cancel{x \leq |\pm(0.25)_{10}|}$ $(-0.25, 0.25)$

Overflow to $\pm\infty$: $\cancel{x \geq |\pm(63)_{10}|}$ $+\infty \rightarrow (63, \infty)$

$-\infty \rightarrow (-\infty, -63)$

Answer to the Question No 2

$2x^2 - 70x + 3 = 0$

$\Rightarrow x^2 - 35x + \frac{3}{2} = 0$

(a) We know,

for quadratic equation, root, $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\Rightarrow x = \frac{-(-35) \pm \sqrt{(-35)^2 - 4 \times 1 \times \frac{3}{2}}}{2 \times 1}$$

$$= \frac{35 \pm \sqrt{1219}}{2}$$

$$= 17.5 \pm \sqrt{304.750}$$

$$\Rightarrow x = 17.5 \pm 17.4571$$

$x_1 = 17.5 + 17.4571 = 34.9571$

$x_2 = 17.5 - 17.4571 = 0.0429000$

(b) $P_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots a_n x^n$

If $x_1, x_2, \cdots x_n$ are the roots, then,

$(x - x_1)(x - x_2) \cdots (x - x_n) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$

Coefficient of x^0: $x_1 x_2 \cdots x_n = a_0$ —— (i)

Coefficient of x^1: $x_2 \cdots x_n + x_1 x_3 \cdots x_n + \cdots + x_1 \cdots x_{n-1} = a_1$ —— (ii)

⋮ ⋮

Coefficient of x^n: $1 = a_n$

For the given equation, $(x - x_1)(x - x_2) = \cancel{a_0}\ \frac{3}{2} + (-35)x + x^2$

$[x^0]$: $x_1 x_2 = \frac{3}{2}$ —— (iii)

$[x^1]$: $-x_2 - x_1 = -35$

$\Rightarrow x_1 + x_2 = 35$ —— (iv)

$[x^2]$: $1 = 1$

~~For~~ (iii) and (iv) equations are the fundamental properties the solution of the given quadratic equation must satisfy.

Checking, (iv) $\Rightarrow x_1 + x_2 = 34.9571 + 0.0429000 = 35.0000$

(iii) $\Rightarrow x_1 x_2 = 34.9571 + 0.0429000 = 1.49966 \approx 1.50000$

∴ My solutions satisfy this properties.

Answer to the Question No 3

Given, $f(x) = \cos(x) + \sin(x)$; $f(0.15) = 1.1382$; $m = 5$

(a) Let. $x_0 = 0$

Taylor series, $$g(x) = \underbrace{g(x_0)}_{a_0} + \underbrace{g'(x_0)}_{a_1} x + \underbrace{\frac{g^2(x_0)}{2!}}_{a_2} x^2 + \underbrace{\frac{g^3(x_0)}{3!}}_{a_3} x^3 + \underbrace{\frac{g^4(x_0)}{4!}}_{a_4} x^4 + \underbrace{\frac{g^5(x_0)}{5!}}_{a_5} x^5 + \cdots$$

For, $$\cos(x) = \cos(0) + (-\sin(0))x + \frac{(-\cos(0))}{2!}x^2 + \frac{\sin(0)}{3!}x^3 + \frac{\cos(0)}{4!}x^4 + \frac{(-\sin(0))}{5!}x^5 + \cdots$$

For $$\sin(x) = \sin(0) + \cos(0)x + \frac{(-\sin(0))}{2!}x^2 + \frac{(-\cos(0))}{3!}x^3 + \frac{\sin(0)}{4!}x^4 + \frac{\cos(0)}{5!}x^5 + \cdots$$

Now,

$$f(x) = \cos(x) + \sin(x)$$

$$= \cos(0) + \sin(0) + ((-\sin(0)) + (\cos(0)))x + ((-\cos(0)) + (-\sin(0)))\frac{1}{2!}x^2$$
$$+ ((\sin(0)) + (-\cos(0)))\frac{1}{3!}x^3 + ((\cos(0)) + (\sin(0)))\frac{1}{4!}x^4$$
$$+ ((-\sin(0)) + (\cos(0)))\frac{1}{5!}x^5 + \cdots$$

$$= \underbrace{1}_{a_0} + \underbrace{1}_{a_1}x + \underbrace{\left(-\frac{1}{2!}\right)}_{a_2}x^2 + \underbrace{\left(-\frac{1}{3!}\right)}_{a_3}x^3 + \underbrace{\frac{1}{4!}}_{a_4}x^4 + \underbrace{\frac{1}{5!}}_{a_5}x^5 + \cdots$$

$$f(x) = 1 + x - \frac{1}{2!}x^2 - \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \frac{1}{5!}x^5 + \cdots$$

Taylor coefficients,

$a_0 = 1$

$a_1 = 1$

$a_2 = -\frac{1}{2!}$

$a_3 = -\frac{1}{3!}$

$a_4 = \frac{1}{4!}$

$a_5 = \frac{1}{5!}$

Given,

(b) Upto 1^{st} term : $f(0.15) = 1$

Upto 2^{nd} term : $f(0.15) = 1 + 0.15 = 1.15$

Upto 3^{rd} term : $f(0.15) = 1 + 0.15 - \frac{1}{2!}(0.15)^2 = 1.13875 \approx 1.1388$

Upto 4^{th} term : $f(0.15) = 1 + 0.15 - \frac{1}{2!}(0.15)^2 - \frac{1}{3!}(0.15)^3 = 1.13818875 \approx 1.1382$

As, 3^{rd} Degree Taylor Polynomial can guarantee ~~8~~ accuracy upto 5 significant figure.

So, the degree of the interpolation polynomial is 3.

(c) We know, Upper bound Error $= \left| \frac{f^{n+1}(\xi)}{(n+1)!} x^{n+1} \right|$

Given, $n = 3$

$x = 0.15$

$$= \underbrace{\left| \frac{f^4(\xi)}{4!} \right|}_{\text{Let. } g} \underbrace{\left| x^4 \right|}_{w(x)}$$

$$g = \frac{f^4(\xi)}{4!} = \frac{1}{4!}(\cos(\xi) + \sin(\xi))$$

$w(x) = x^4$

$w(x)_{max} = (0.15)^4 = 0.00050625$

$$g_{max} = \frac{1}{4!}(\cos(0.15) + \sin(0.15))$$

$$= \frac{1}{4!}(0.98877 + 0.14944)$$

$\therefore g_{max} = 0.047425$

$\therefore$ Upper bound of the interpolation Error $= | g_{max} * w(x)_{max} |$

$$= |0.047425 * 0.00050625|$$

$$= 0.000024009$$

$$= \cancel{24}\ 0.24009 \times 10^{-4}$$

Answer to the Question No 4

Given, $f(x) = \sin(x) + \cos(x)$ at the nodes $\{-\pi, -\pi/2, 0\}$

x	$f(x)$
$x_0 = -\pi$	$-1 = f(x_0)$
$x_1 = -\frac{\pi}{2}$	$-1 = f(x_1)$
$x_2 = 0$	$1 = f(x_2)$

Total Node $= n+1 = 3$

Degree $= n = 2$

(a) $P_2(x) = a_0 x^0 + a_1 x^1 + a_2 x^2$

$$P_2(-\pi) = a_0 + a_1(-\pi) + a_2\pi^2 = -1$$

$$P_2\left(-\frac{\pi}{2}\right) = a_0 + a_1\left(-\frac{\pi}{2}\right) + a_2\frac{\pi^2}{4} = -1$$

$$P_2(0) = a_0 + a_1 \cdot 0 + a_2 \cdot 0 = 1$$

Using Vandermond Matrix

$$\begin{bmatrix} 1 & -\pi & \pi^2 \\ 1 & -\pi/2 & \frac{\pi^2}{4} \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \quad V \cdot A = Y$$

$$\Rightarrow A = V^{-1}Y$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1/\pi & -4/\pi & 3/\pi \\ 2/\pi^2 & -4/\pi^2 & 2/\pi^2 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -\frac{1}{\pi} + \frac{4}{\pi} + \frac{3}{\pi} \\ -\frac{2}{\pi^2} + \frac{4}{\pi^2} + \frac{2}{\pi^2} \end{bmatrix}$$

$$\therefore \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 6/\pi \\ 4/\pi^2 \end{bmatrix} \quad ; \quad a_0 = 1, a_1 = \frac{6}{\pi}, \; a_2 = \frac{4}{\pi^2}$$

$\therefore$ Interpolation Polynomial,

$$P_2(x) = 1 + \frac{6}{\pi}x + \frac{4}{\pi^2}x^2$$

Vandermond Matrix,

$$V = \begin{bmatrix} 1 & -\pi & \pi^2 \\ 1 & -\pi/2 & \pi^2/4 \\ 1 & 0 & 0 \end{bmatrix}$$

Now: $\det(V) = 1(0-0) + \pi\left(-\frac{\pi^2}{4}\right) + \pi^2\left(\frac{\pi}{2}\right.$

$$= \frac{\pi^3}{4}$$

Cofactor matrix,

$$C = \begin{bmatrix} 0 & \frac{\pi^2}{4} & \frac{\pi}{2} \\ 0 & -\pi^2 & -\pi \\ \frac{\pi^3}{4} & \frac{3\pi^2}{4} & \frac{\pi}{2} \end{bmatrix}$$

$$V^{-1} = \frac{1}{\det(V)} \text{Adj}(V) = \frac{1}{\frac{\pi^3}{4}} C^T$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 1/\pi & -4/\pi & 3/\pi \\ 2/\pi^2 & -4/\pi^2 & 2/\pi^2 \end{bmatrix}$$

(b) Using Lagrange theorem,

$$P_2(x) = l_0(x)\,f(x_0) + l_1(x)\,f(x_1) + l_2(x)\,f(x_2)$$

Lagrange Bases,

$$l_0(x) = \frac{x-x_1}{x_0-x_1}\cdot\frac{x-x_2}{x_0-x_2} = \frac{x+\frac{\pi}{2}}{-\pi+\frac{\pi}{2}}\cdot\frac{x-0}{-\pi-0} = \frac{2}{\pi^2}\left(x+\frac{\pi}{2}\right)(x)$$

$$l_1(x) = \frac{x-x_0}{x_1-x_0}\cdot\frac{x-x_2}{x_1-x_2} = \frac{x+\pi}{-\frac{\pi}{2}+\pi}\cdot\frac{x-0}{-\frac{\pi}{2}-0} = \frac{-4}{\pi^2}(x+\pi)(x)$$

$$l_2(x) = \frac{x-x_0}{x_2-x_0}\cdot\frac{x-x_1}{x_2-x_1} = \frac{x+\pi}{0+\pi}\cdot\frac{x+\frac{\pi}{2}}{0+\frac{\pi}{2}} = \frac{2}{\pi^2}(x+\pi)\left(x+\frac{\pi}{2}\right)$$

Lagrange interpolation polynomial,

$$P_2(x) = \frac{2}{\pi^2}\left(x+\frac{\pi}{2}\right)(x)(-1) + \frac{-4}{\pi^2}(x+\pi)(x)(-1) + \frac{2}{\pi^2}(x+\pi)\left(x+\frac{\pi}{2}\right)$$

$$= -\frac{2}{\pi^2}x^2 - \frac{1}{\pi}x + \frac{4}{\pi^2}x^2 + \frac{4}{\pi}x + \frac{2}{\pi^2}x^2 + \frac{2}{\pi}x + \frac{1}{\pi}x + 1$$

$$\therefore\ P_2(x) = 1 + \frac{6}{\pi}x + \frac{4}{\pi^2}x^2$$